

Week 2 Project Summary

Project Title

Interactive Netflix Data Visualization Dashboard using Streamlit

Objective

The goal of this project was to analyze the **Netflix Movies and TV Shows** dataset and create an interactive dashboard with **Streamlit**. The dashboard lets users explore the dataset through filters, charts, KPI cards, and downloadable results, making data analysis straightforward and informative.

Dataset Information

Dataset Name: Netflix Movies and TV Shows Dataset

Source: Kaggle

Total Records: 8,807

Total Columns: 12

This dataset includes information about Netflix titles, such as content type, release year, country, rating, date added, duration, and other features.

Data Preparation

Before building the dashboard, several steps were taken to ensure data quality:

- Removed duplicate records.
- Addressed missing values.
- Changed the *date_added* column to *datetime* format.
- Extracted the *Year Added* feature.
- Cleaned and organized the dataset for visualization.

Dashboard Features

1. Project Introduction

This section introduces the Netflix Data Visualization Dashboard and explains its purpose. The dashboard is made using **Python, Pandas, NumPy, Matplotlib, Seaborn, Plotly, and Streamlit**.

2. Dataset Overview

This part displays general information about the dataset, including:

- Total Rows: **8,807**
- Total Columns: **12**

- This gives users a quick view of the dataset's size.

3. Dataset Preview

This shows the first few rows of the dataset, helping users understand the available columns and data structure.

4. Interactive Sidebar Filters

The dashboard features interactive filters that let users explore the dataset by:

- Content Type (Movie or TV Show)
- Country

All visualizations and KPI cards refresh automatically with the selected filters.

5. KPI Cards

The dashboard shows key performance indicators, including:

- Total Records
- Total Movies
- Total TV Shows
- Total Countries

These KPIs offer a quick summary of the filtered dataset.

6. Interactive Visualizations

The dashboard includes various charts to analyze Netflix content in different ways:

- Movies vs TV Shows (Bar Chart)
- Content Rating Distribution (Pie Chart)
- Content Added Over Years (Line Chart)
- Release Year Distribution (Histogram)
- Release Year by Type (Box Plot)
- Top 10 Countries by Content (Bar Chart)
- Release Year vs Content Type (Scatter Plot)

These visualizations help users spot trends, distributions, and comparisons in the dataset.

7. Correlation Heatmap

A correlation heatmap is included to look at relationships between numerical variables, such as:

- Release Year
- Year Added

This helps determine whether the variables are positively or negatively correlated.

8. Insights and Conclusions

The dashboard automatically provides useful insights based on the dataset, including:

1. Movies make up a larger part of Netflix's catalog than TV Shows.
2. The majority of Netflix content was added between **2017 and 2020**.
3. **TV-MA** is the most common content rating.
4. The **United States** has the highest number of Netflix titles.
5. Most Netflix titles were released after **2010**.
6. The correlation between Release Year and Year Added is weakly positive.

9. Download Filtered Dataset

Users can download the filtered dataset in CSV format for further analysis. This feature makes the dashboard more interactive and useful.

Technologies Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Plotly
- Streamlit

Project Outcome

The project successfully created an **Interactive Netflix Data Visualization Dashboard** using Streamlit. It showcases practical skills in **data preprocessing, exploratory data analysis (EDA), interactive visualization, dashboard development, and data storytelling**. The dashboard allows users to explore Netflix data efficiently through filters, KPI cards, various charts, a correlation heatmap, meaningful insights, and downloadable filtered datasets. This provides an effective solution for interactive data analysis.